

can we reason about the world?

a short history of theory — and where scale runs out

the bet

- the whole enterprise rests on one wager:
- the world is **legible** because it is **mathematical**.
- everything after is footnotes to that bet.

Galileo

- *the book of nature is written in mathematics.*
- nature is legible, not capricious.

Newton — unification I

- heaven + earth -> one law of gravity.
- same physics here and on Jupiter.
- Kelvin-Helmholtz: clouds in our sky **and** Jupiter's bands.

Maxwell — unification II

- electricity + magnetism + light -> one field.
- predicted c from two bench constants.
- purest "we can reason about the world" moment in physics.
- continues: electroweak -> Standard Model -> (gravity: open).

Laplace's demon

- know every position + momentum -> predict everything.
- determinism as the dream of total knowledge.
- note: Laplace sits on **both** sides of this story.

the demon retreats

- Maxwell + Boltzmann (1860s-70s): statistical mechanics.
- stop tracking every particle. reason about ensembles.
- temperature = ignorance, managed well.
- physics **chooses** statistics over the demon.

uncertainty gets a calculus

- Pascal-Fermat, 1654: the problem of points.
- the "probabilistic revolution" begins.
- Bayes / Laplace: inverse probability = learning from evidence.
- ancestor of everything ML.

the cracks

- Poincare (~1890): 3 bodies -> determinism $\neq$ predictability.
- quantum: irreducible chance.
- Godel: truths no proof reaches.
- Turing (1936): undecidability **and** the birth of computation.

the computational turn

- Shannon (1948): uncertainty gets a unit — the bit.
- Monte Carlo (Ulam/von Neumann): compute by sampling.
- Lorenz (1963): chaos found **on a computer**.
- von Neumann: rationality as empirical science. fire up the sim.

where we are now

- Kolmogorov / Solomonoff: shortest program = best explanation.
- Pearl: causality — reason about interventions, not correlations.
- Gigerenzer: a **second** revolution — the bounded mind, a toolbox of fast & frugal heuristics.

the modern claim

- Sutton, *The Bitter Lesson* (2019):
- general methods that **leverage computation** always win.
- cleverness loses to scale + search.
- ... does it?

the hidden precondition

- search and learning both **spend** evaluations.
- the lesson assumes a **free oracle**: cheap, instant, ~infinite.
- chess/Go: rules give exact win/loss, unlimited self-play.
- speech/vision: labeled corpora, a loss in microseconds.

slow properties have no free oracle

- maintainability: signal months later, noisy, can't self-play.
- reliability: tail events — needs long runtime or proof.
- human understanding: reward = a person gets a testable idea.
- each eval is **slow, costly, human**.

the real axis

- not scale vs cleverness.
- **eval cost x latency x availability.**
- cheap/instant evals -> Sutton wins.
- expensive/slow/human evals -> frugal is the **only** thing that runs.

biting the future

- the lesson is a margin loan on Moore's law.
- assumes resources keep growing exponentially.
- Dennard dead (~2006). cost/transistor flat past 28nm.
- peak data. megawatt training. returns are power-law.
- exponential -> **logistic**. the collateral is capping.

Wirth's witness

- reflex: "write it loose, faster CPUs bail you out."
- Wirth's law: bloat **outruns** hardware. you fall behind.
- the bet lost **during** the boom — exponential fully running.
- Wirth is the prosecution, not the defense.

Jevons closes the door

- Jevons (1865): cheaper coal -> **more** coal burned.
- cheapen a resource, demand eats the surplus.
- cheaper compute -> bigger models -> nothing banked.
- frugality is **chosen**, never delivered.

debloat

- every dependency = a margin loan on future **attention**.
- cheap to add (compiles, tests green) — slow to pay (CVEs, churn).
- left-pad / Log4Shell / xz: dep count is now measured risk.
- not "fewer deps" — fewer **un-evaluated, low-trust** deps.
- eval-weighted minimalism.

the thesis

- scale beats cleverness only where evaluation is free.
- most of software has **no free oracle**.
- bounded resource, bounded mind, frugal toolbox.
- *simple ain't stupid*.

close

- Gigerenzer's second revolution, replayed in SE.
- the omniscient demon was always a dream.
- build for the world we're in: bounded, capping, costly to check.